F=ma Re-entry Diagnostic

Fillable PDF - 20 questions - 60 minutes

Use g = 10 m/s^2. No notes or formula sheet.

Student information

Name:

Date: Start time: End:

Confidence key

C = confident U = unsure G = guessed

Before you begin

- Work independently and do not check answers during the test.
- Select one answer and one confidence level for each question.
- Use the large work boxes to show your setup and reasoning.
- Flag any question that takes more than 3 minutes.

Questions taking more than 3 minutes

Setups I could not visualize

1. Average Speed

A cyclist travels 12 km at 6 m/s and returns along the same route at 4 m/s. What is the average speed?

- [ ] A. 4.0 m/s
- [ ] B. 4.8 m/s
- [ ] C. 5.0 m/s
- [ ] D. 5.2 m/s
- [ ] E. 6.0 m/s

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

2. Same-Height Motion

A ball is thrown vertically upward. It passes a balcony at t = 2 s and again at t = 8 s. What is its initial speed?

- [ ] A. 20 m/s
- [ ] B. 30 m/s
- [ ] C. 40 m/s
- [ ] D. 50 m/s
- [ ] E. 60 m/s

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

3. Exact Requested Quantity

Using Question 2, what is the balcony's height above the launch point?

- ☐ A. 40 m
- ☐ B. 60 m
- ☐ C. 80 m
- ☐ D. 100 m
- ☐ E. 125 m

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

4. Braking

A car can stop from speed v in distance d. Under the same constant braking force, what distance is needed from speed 1.5v?

- ☐ A. 1.5d
- ☐ B. 2d
- ☐ C. 2.25d
- ☐ D. 2.5d
- ☐ E. 3d

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

5. Two-Road Braking

A car would stop in 120 m on dry road. It travels 30 m on dry road and then reaches ice, where friction is one-third as large. What is the total stopping distance?

- ☐ A. 210 m
- ☐ B. 270 m
- ☐ C. 300 m
- ☐ D. 330 m
- ☐ E. 390 m

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

6. Falling-Time Trap

A stone falls from rest from height H. After half of its total fall time, what fraction of H remains below it?

- ☐ A. 1/4
- ☐ B. 1/2
- ☐ C. 2/3
- ☐ D. 3/4
- ☐ E. 7/8

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

7. Projectile

A ball is launched horizontally at 20 m/s from a 45 m cliff. How far from the base does it land?

- ☐ A. 30 m
- ☐ B. 40 m
- ☐ C. 60 m
- ☐ D. 90 m
- ☐ E. 120 m

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

8. Velocity Versus Position

A particle starts from rest under constant acceleration. If its speed is v after traveling distance x, what is its speed after traveling distance 4x?

- ☐ A. v
- ☐ B. sqrt(2)v
- ☐ C. 2v
- ☐ D. 4v
- ☐ E. 16v

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

9. Elevator Scale

A 50 kg person stands in an elevator accelerating downward at 3 m/s^2. What does the scale read?

- ☐ A. 150 N
- ☐ B. 350 N
- ☐ C. 500 N
- ☐ D. 650 N
- ☐ E. 800 N

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

10. Angled Pull and Friction

A block is pulled across a horizontal floor by force F directed at angle theta above horizontal. What is the kinetic friction magnitude?

- ☐ A. mu_k mg
- ☐ B. mu_k(mg + F sin theta)
- ☐ C. mu_k(mg - F sin theta)
- ☐ D. mu_k F cos theta
- ☐ E. mu_k F sin theta

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

11. Static Friction

A block rests on an incline at angle theta. What minimum coefficient of static friction prevents slipping?

- ☐ A. sin theta
- ☐ B. cos theta
- ☐ C. tan theta
- ☐ D. cot theta
- ☐ E. 1

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

12. Contact Force

Blocks m and 3m touch on a frictionless floor. Force F is applied to the m block so that it pushes the 3m block. What is the contact force?

- ☐ A. F/4
- ☐ B. F/2
- ☐ C. 3F/4
- ☐ D. F
- ☐ E. 4F/3

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

13. Atwood Machine

Masses 4m and m hang from an ideal pulley. What is their acceleration magnitude?

- ☐ A. g/5
- ☐ B. 2g/5
- ☐ C. 3g/5
- ☐ D. 4g/5
- ☐ E. g

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

14. Atwood Tension

Using Question 13, what is the rope tension?

- ☐ A. 2mg/5
- ☐ B. 4mg/5
- ☐ C. 6mg/5
- ☐ D. 8mg/5
- ☐ E. 2mg

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

15. Scale Supporting Pulley

The pulley in Question 13 hangs from a scale. What does the scale read?

- ☐ A. 8mg/5
- ☐ B. 2mg
- ☐ C. 12mg/5
- ☐ D. 16mg/5
- ☐ E. 5mg

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

16. Movable Pulley Constraint

A movable pulley rises 4 cm. By how far does the free end of its supporting rope move?

- ☐ A. 2 cm
- ☐ B. 4 cm
- ☐ C. 6 cm
- ☐ D. 8 cm
- ☐ E. 16 cm

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

17. Top of a Hill

A car travels over the top of a circular hill. As its speed increases, the normal force from the road:

- ☐ A. increases
- ☐ B. remains mg
- ☐ C. decreases
- ☐ D. reverses direction
- ☐ E. becomes mv^2/R

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

18. Loss of Contact

A car passes over a circular hill of radius R. At what speed does it just lose contact?

- ☐ A. sqrt(gR)/2
- ☐ B. sqrt(gR)
- ☐ C. sqrt(2gR)
- ☐ D. 2sqrt(gR)
- ☐ E. gR

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

19. Banked Curve

A frictionless curve is banked at angle theta. Which relation is correct?

- ☐ A. v^2 = gR sin theta
- ☐ B. v^2 = gR cos theta
- ☐ C. v^2 = gR tan theta
- ☐ D. v^2 = gR / tan theta
- ☐ E. v = gR tan theta

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

20. Conical Pendulum

A conical pendulum's string makes angle theta with the vertical. What is its tension?

- ☐ A. mg sin theta
- ☐ B. mg cos theta
- ☐ C. mg tan theta
- ☐ D. mg / sin theta
- ☐ E. mg / cos theta

Confidence: ☐ C ☐ U ☐ G **Over 3 min:** ☐

Work / reasoning:

Final Answer Summary

Q	Answer	Confidence	Q	Answer	Confidence
1			11		
2			12		
3			13		
4			14		
5			15		
6			16		
7			17		
8			18		
9			19		
10			20		

Time remaining:

Strongest topics

Weakest topics

Conceptual mistakes or setups to review